

Classification of Algebraic Varieties II

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Outline of the talk

1 Introduction

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- 2 The MMP for 3-folds

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Notation

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- K_X denotes (a choice of) the **canonical divisor** i.e. a formal linear combination of codimension 1 subvarieties of X such that $\omega_X = \mathcal{O}_X(K_X)$.
- We would like to use ω_X (or equivalently K_X) to understand the geometry of X .

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is defined by $x \rightarrow [s_0(x) : \dots : s_N(x)]$. (This map is not defined on the common zeroes of $s_0, \dots, s_N$.)

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- If X and X' are birational, then $\forall m \geq 0$

$$H^0(X, \omega_X^{\otimes m}) \cong H^0(X', \omega_{X'}^{\otimes m}).$$

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- If $\kappa(X) = \dim(X)$, we say that X is of **general type**.
- One would like to understand the structure of this ring and to use its features to classify complex projective varieties.

Finite generation

Today I would like to discuss the following result:

Theorem (Birkar-Cascini-Hacon-M^cKernan and Siu)

Let X be a smooth projective variety, then $R(\omega_X)$ is finitely generated.

The two proofs are independent. Our proof is algebraic and uses the ideas of the MMP (minimal model program). Siu's proof is analytic and requires X to be of general type.

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- 1 Introduction
- 2 The MMP for 3-folds
 - Threefolds
 - Running the minimal model program
- 3 Higher dimensional MMP

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Theorem

Assume $\dim(X) = 3$. If $\kappa(X) < 0$, then X is covered by rational curves ($\mathbb{P}^1$'s) and hence $R(\omega_X) \cong \mathbb{C}$.

If $\kappa(X) \geq 0$, then there is a birational map $X \dashrightarrow X_{\min}$ such that $\omega_{X_{\min}}$ is nef (i.e. $\omega_{X_{\min}} \cdot C \geq 0$ for any curve $C \subset X_{\min}$).

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- More precisely, if $\kappa(X) \geq 0$, there is a sequence of well understood birational maps (flips and divisorial contractions)

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_M = X_{\min}.$$

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 - 2 The varieties X_i have mild singularities.

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- There are several new features:
 - ① The minimal model X_{min} is not unique.
 - ② The varieties X_i have mild singularities.
- It is however known that $\omega_{X_{min}}$ is semiample (i.e. $\phi_{\omega_{X_{min}}^{\otimes m}} : X \rightarrow \mathbb{P}_{\mathbb{C}}^N$ is a morphism) and so $\omega_{X_{min}}^{\otimes m} = \phi_{\omega_{X_{min}}^{\otimes m}}^* \mathcal{O}_{\mathbb{P}_{\mathbb{C}}^N}(1)$. In particular it follows that $R(\omega_X)$ is finitely generated.

Running a MMP

- A **Minimal Model program** is a sequence of geometric operations $X_i \dashrightarrow X_{i+1}$ that starting from a projective variety X produces a minimal model (if $\kappa(X) \geq 0$) or a covering family of rational curves (if $\kappa(X) < 0$).

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- We start with a projective variety X with mild singularities.
- If ω_X is nef (i.e. $\omega_X \cdot C \geq 0$) we are done ($X = X_{min}$).
- If ω_X is not nef, then consider the **cone of effective curves**

$$\overline{NE}(X) = \left\{ \sum r_i C_i \mid r_i \in \mathbb{R}^{\geq 0} \right\} / \equiv$$

(here $C \equiv C'$ or $[C] = [C']$ if for any line bundle L on X , we have $L \cdot (C - C') = 0$).

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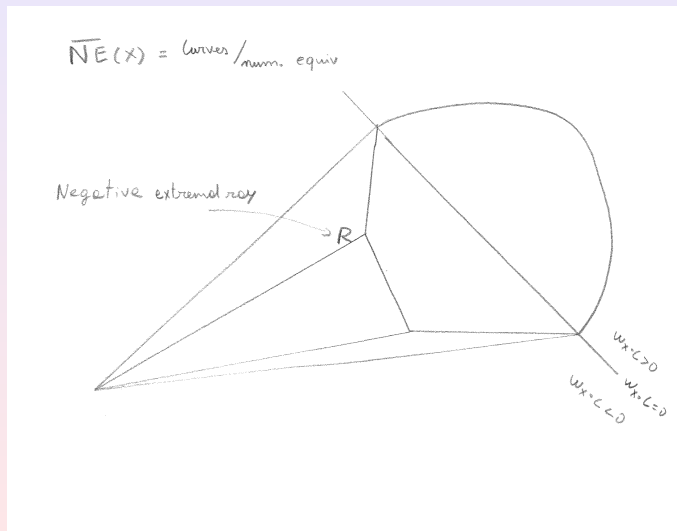
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- By the Cone Theorem, there is a negative extremal ray $R = \mathbb{R}^+[C]$ where $C \subset X$ is a curve such that $\omega_X \cdot C < 0$ and there is a morphism $\text{Cont}_R : X \rightarrow Z$ such that a curve $D \subset X$ is contracted if and only if $[D] \in R$.

The Cone Theorem



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- If $\dim Z = \dim X$ and Cont_R is **small** (i.e. $\dim \text{Exc}(X \rightarrow Z) \leq \dim X - 2$) then Z has “bad” singularities.
- Instead of replacing X by Z , we will replace X by its flip X^+ .

Flips

- The **flip** of $X \rightarrow Z$ is a birational morphism $X^+ \rightarrow Z$ s. t.
 - 1) $X \dashrightarrow X^+$ is an isomorphism on the complement of $\text{Exc}(X \rightarrow Z)$,
 - 2) $\dim \text{Exc}(X^+ \rightarrow Z) \leq \dim(X) - 2$,
 - 3) X^+ has mild singularities, and
 - 4) If $C \subset \text{Exc}(X^+ \rightarrow Z)$, then $\omega_{X^+} \cdot C > 0$.

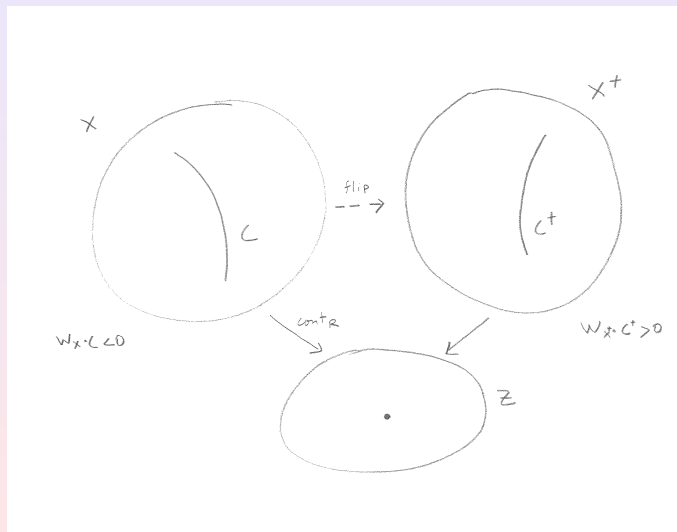
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- In other words, the flip $X \dashrightarrow X^+$ is a $\text{codim} \geq 2$ surgery that replaces ω_X negative curves by ω_{X^+} positive curves.
- If the flip of $X \rightarrow Z$ exists, it is unique and given by $\text{Proj}_Z R(\omega_X)$. So that to construct the flip we must show that $R(\omega_X)$ is finitely generated over Z (i.e. if $Z = \text{Spec} A$, $R(\omega_X)$ is a finitely generated A -algebra).

Picture of a flip



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- In order to complete the MMP, one must show that there are no infinite sequences of flips and divisorial contractions.
- If $X \rightarrow Z$ is a divisorial contraction, then $\rho(Z) = \rho(X) - 1 \geq 0$. If $X \dashrightarrow X^+$ is a flip, then $\rho(X^+) = \rho(X)$. Therefore, it suffices to show that there is no infinite sequence of flips.

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- In dimension 3, this is relatively easy to do.

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 - Reduction to log pairs of general type
 - Idea of the proof

Log pairs

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- The proof proceeds by induction on the dimension of X . We relate ω_X (equivalently K_X) to its restriction to a divisor S (i.e. a codimension 1 subvariety which typically belongs to $|mK_X|$).
- This is achieved by the adjunction formula which has the form $(K_X + S)|_S = K_S + \Delta_S$ for some $\mathbb{Q}$ -divisor Δ_S .

A Theorem of Fujino-Mori

Mori and Fujino have shown that:

Theorem

Let (X, Δ) be any KLT $\mathbb{Q}$ -factorial pair with $\Delta \in \text{Div}_{\mathbb{Q}}(X)$. Then there exists a KLT $\mathbb{Q}$ -factorial pair (Y, Γ) of general type with $\Gamma \in \text{Div}_{\mathbb{Q}}(Y)$ such that

$$R(K_X + \Delta)^{(m)} \cong R(K_Y + \Gamma)^{(m')}$$

for some integers $m, m' > 0$.

Note that

$$\kappa(K_X + \Delta) = \kappa(K_Y + \Gamma) = \dim Y.$$

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- If (X, Δ) is KLT and $K_X + \Delta$ is nef and of general type, then by the Base Point Free Theorem of Kawamata and Shokurov, $K_X + \Delta$ is semiample.
- As we have mentioned before, this implies that $R(K_X + \Delta)$ is finitely generated.
- Therefore, to show that $R(\omega_X)$ is finitely generated (for any smooth complex projective variety), it suffices to show that minimal models exist for KLT pairs of general type.

Existence of minimal models for KLT pairs of general type.

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let (X, Δ) be any KLT pair of general type. Then there is a minimal model $f : X \dashrightarrow X_{\min}$ (i.e. $K_{X_{\min}} + f_\Delta$ is semiample).*

Corollary

Flips exist (for KLT pairs; in any dimension).

Remark: We are unable to show termination of flips. But we can pick a sequence of flips that terminates.

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- (Birkar-Cascini-Hacon-M^cKernan) If flips exist in dimension $n + 1$ (and minimal models exist in dimension n) then minimal models exist in dimension $n + 1$. (Uses the ideas of the MMP and in particular techniques developed by Shokurov.)

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- By adjunction, we may write $(K_X + \Delta)|_S = K_S + \Delta_S$ where (S, Δ_S) is KLT.
- Shokurov's observation is that to show that $R(K_X + \Delta)$ is finitely generated over Z (and hence that the flip exists), it suffices to show that the restricted algebra

$$R_S(K_X + \Delta) := \text{Im}(R(K_X + \Delta) \rightarrow R(K_S + \Delta_S))$$

is finitely generated (over Z).

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- The divisor Θ is the limit of $\mathbb{Q}$ -divisors Θ_m defined by subtracting common components of Δ_S and $\text{Bs}(m(K_X + \Delta))/m$ from Δ_S .

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- Using a Diophantine approximation argument, we show that Θ is in fact a $\mathbb{Q}$ -divisor and $\Theta = \Theta_m$ for some $m > 0$.

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- In this way, we obtain a sequence of rational numbers $1 \geq t_1 \geq t_2 \cdots \geq 0$ and of flips/div. contractions $X_i \dashrightarrow X_{i+1}$ such that X_i is a minimal model for $(X, \Delta + t_i A)$.

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- This implies that the sequence $\dots X_i \dashrightarrow X_{i+1} \dots$ is not infinite and so the above process produces the required minimal model.
- Note that to prove the above result on the finiteness of minimal models, we must use a compactness argument and hence it is critical to work with $\mathbb{R}$ -divisors.